

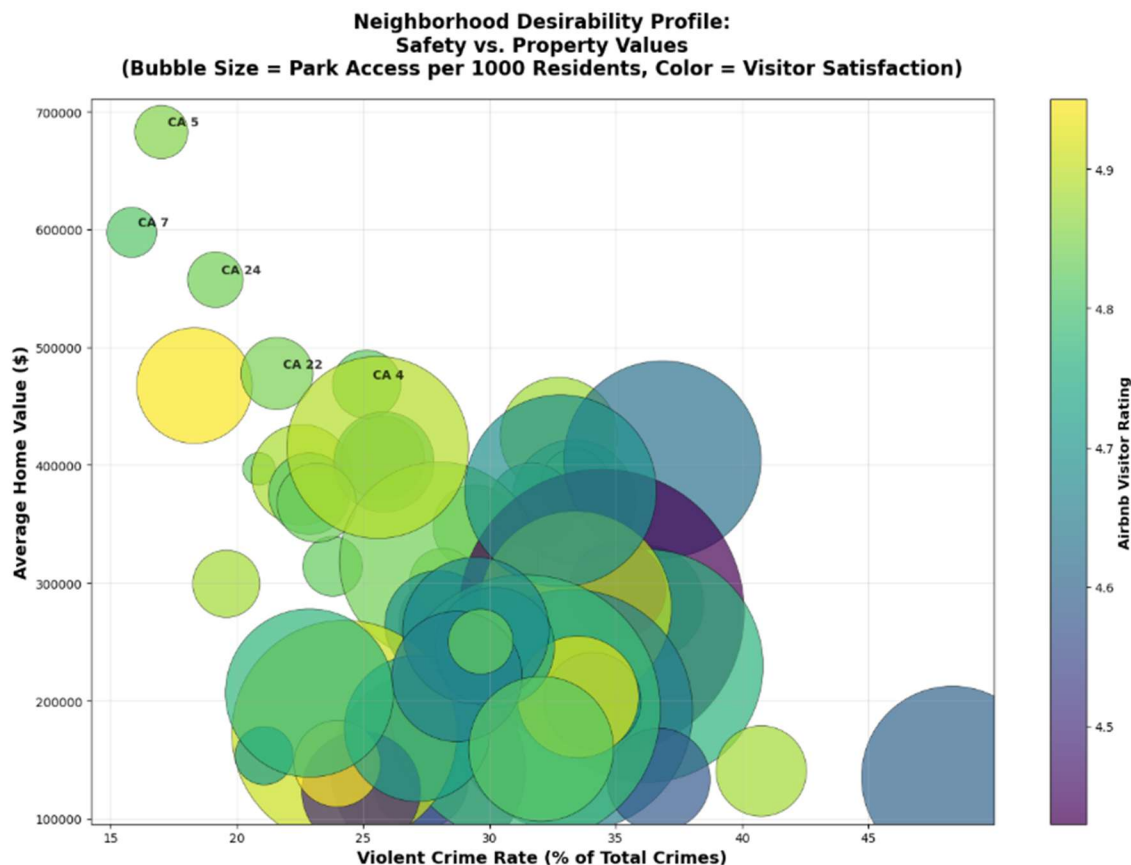
Chicago Neighborhood Desirability Analysis

Introduction

This project examines what drives neighborhood desirability in Chicago, proxied by the Zillow Home Value Index (ZHVI), which reflects how markets price safety, amenities, and economic opportunity. The goal was to identify which neighborhood characteristics most influence residential property values across 53 Chicago community areas and determine which machine learning model provides the most actionable insights for city planners.

Data & Methodology

I used data from public sources, including Zillow home values (2000–2024), crime statistics (through 2025), food inspections, parks, and Airbnb listings (2023–2025). Predictor variables included crime metrics (total crimes, violent crimes, violent crime rate), dining variables (restaurant count, high-risk restaurants, food safety ratios), recreational facilities (park size), service demand (311 calls and density), Airbnb data (listing count, average price, occupancy, revenue, ratings, market penetration, revenue efficiency), and past home value growth. The final dataset included 53 community areas, with 39 (74%) used for training and 14 (26%) for testing. I tested two models: a Random Forest Regressor and an XGBoost Gradient Boosting Model. Performance was evaluated using R-squared, mean absolute error (MAE), and root mean squared error (RMSE).



Results & Analysis

XGBoost outperformed Random Forest. On the test set, XGBoost explained 71% of home value variance ($R^2 = 0.71$) with an MAE of \$65,574 and RMSE of \$83,250, and only a 6.5% train-test gap, indicating strong generalization on this split. Random Forest explained only 32% of test variance ($R^2 = 0.32$) with an MAE of \$109,519 and a 17% train-test gap, showing weaker predictive performance.

Feature importance from XGBoost shows that average Airbnb rating is the strongest predictor, followed by historical home value growth, high-risk business density, and violent crime rate. Parks, service requests, and Airbnb occupancy matter less but still contribute. This suggests that market performance, safety, and commercial environment are the main drivers of neighborhood value in Chicago.

A limitation I found is that 5-fold cross-validation on the training set gives an average R^2 of only ~ 0.21 with a high standard deviation (~ 0.23). This means performance is sensitive to which community areas are held out. These results are best interpreted as patterns and relative feature importance rather than perfectly stable predictions.

Model Comparison & Discussion

Random Forest is easier to interpret but underperforms, leaving much neighborhood variation unexplained. XGBoost captures non-linear effects and interactions, such as the amplified impact of violent crime, the combined effect of high-risk businesses, and the influence of Airbnb performance. While less interpretable, XGBoost's higher R^2 and small train-test gap make it the preferred tool for city planners. The CV results serve as a reminder to be cautious in expecting consistent numeric predictions across neighborhoods.

Conclusion & Future Steps

XGBoost predicts Chicago home values with R^2 of 0.71 on the test set and an average error of roughly \$65,574. Airbnb performance, historical home values, commercial density, and violent crime are the primary determinants of neighborhood desirability. City planners can use these insights to monitor Airbnb trends, prioritize crime reduction, and manage commercial development.

The cross-validation results highlight that the model's performance can vary substantially depending on which neighborhoods are held out, so conclusions should focus on general patterns rather than precise numeric predictions. Future work could add features like transit access, school quality, walkability, and demographics to improve accuracy. Testing in other cities and incorporating equity considerations could help make planning fairer and more generalizable.